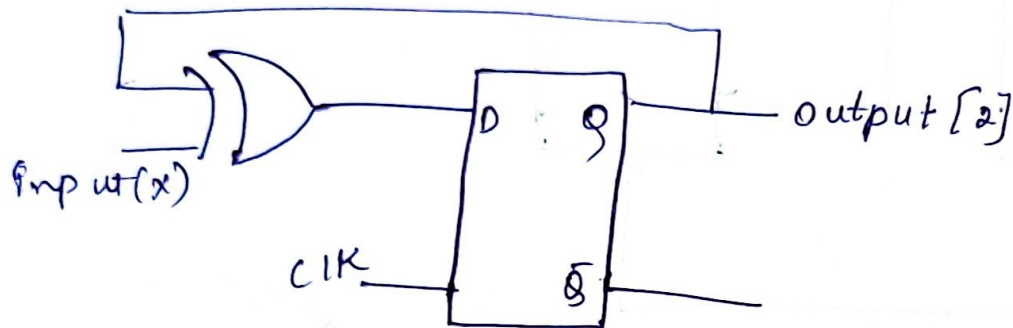


Assignment 4.

1. Explain the behaviour of given circuit



XOR gate inputs are
external $q/p = x$
Feedback $q/p = Q$

$$D = x \oplus Q$$

$$Q_{next} = D$$

$$Q_{next} = x \oplus Q$$

$$Z_{next} = x \oplus Z$$

Case 1 : $x = 0$ $Q_{next} = Q$

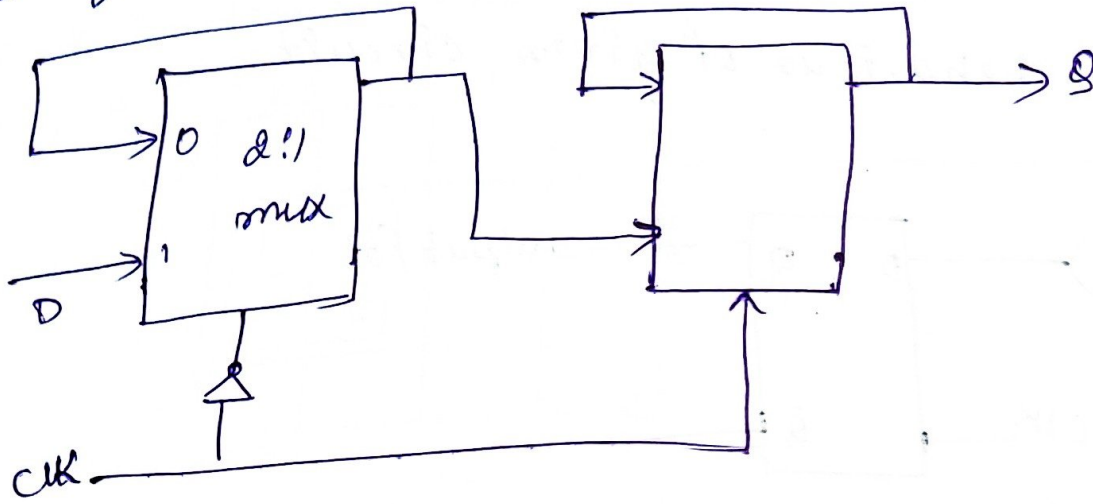
Case 2 : $x = 1$ $Q_{next} = \bar{Q}$

Circuit is TFF

State table

Present (Q)	Input (x)	Q_{next}
0	0	0
0	1	1
1	0	1
1	1	0

2. Design a posedge triggered DFF using 2:1 mux only



$$Q_{next} = CLK \cdot 0 + \overline{CLK} \cdot Q$$

Operation:

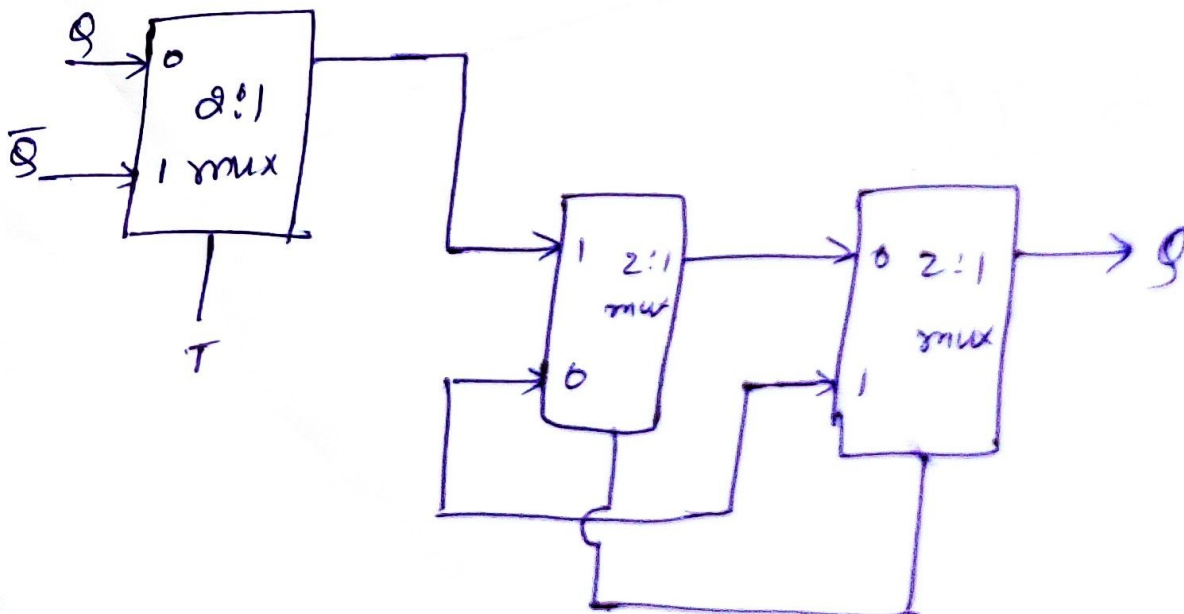
master enable = $\overline{CLK}$

slave enable = CLK

$CLK=0$ master tracks Q/p , slave holds

$CLK=1$ master slaves copies master's output

3. Design a negedge triggered TFF using 2:1 mux



4. Convert by adding external gates

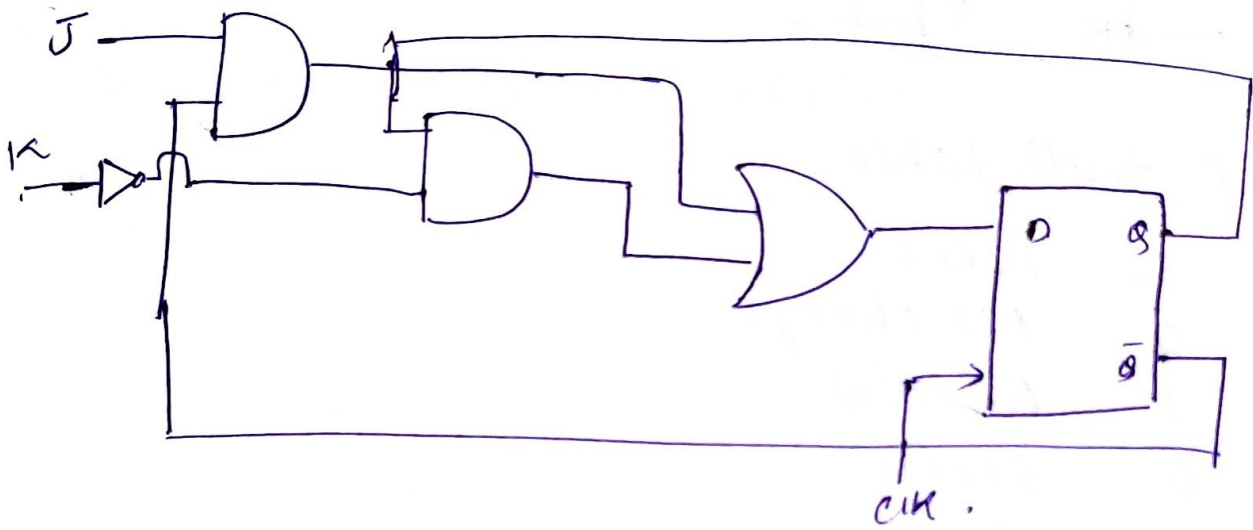
a) A DFF to JKFF

Characteristic equation of JK

$$Q_{n+1} = J\bar{Q} + KQ$$

For DFF $D = Q_{n+1}$

$$\therefore D = J\bar{Q} + \bar{K}Q.$$



b) TFF to DFF

Characteristic eqn of TFF

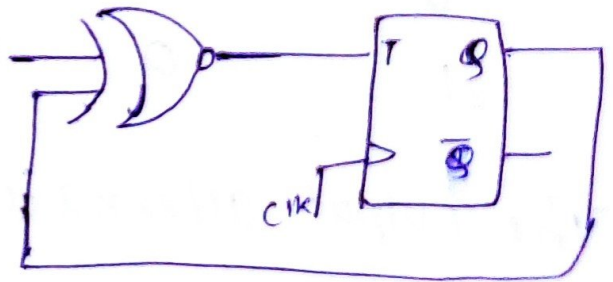
$$.g_{n+1} = T \oplus g$$

For DFF $Q_{n+1} = D$

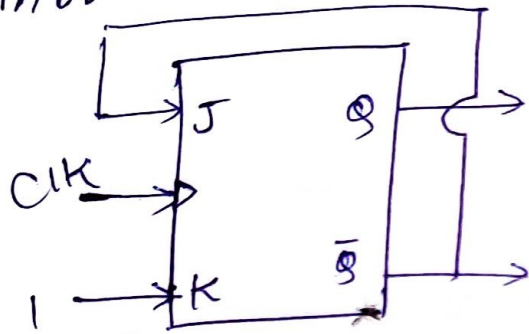
$F_{00} \text{ DFF } Q_{nt+1} = D$

$$D = T \oplus g.$$

$T = D + Q$



5. In a positive edge triggered JKFF, we have $J = Q'$ and $K = 1$ as shown below. Assume the flipflop was initially cleared and then clocked for 6 clock pulses, find the sequence at the Q output for those 6 clock cycles



JKFF truth table

J	K	Next
0	0	No change
0	1	Reset (0)
1	0	Set (1)
1	1	Toggle

Since $J = \bar{Q}$ and $K = 1$

Present Q	$J = \bar{Q}$	K	Action	Next Q
0	1	1	Toggle	1
1	0	1	Reset	0

The output alternates b/w 0 and 1

Positive edge	J	K	Q	$\bar{Q}$
1 st	1	1	1	0
2 nd	0	1	0	1
3 rd	1	1	1	0
4 th	0	1	0	1
5 th	1	1	1	0
6 th	0	1	0	1

The Sequence $Q = 1, 0, 1, 0, 1, 0$